PE43711

Product Specification

UltraCMOS® RF Digital Step Attenuator, 9 kHz–6 GHz

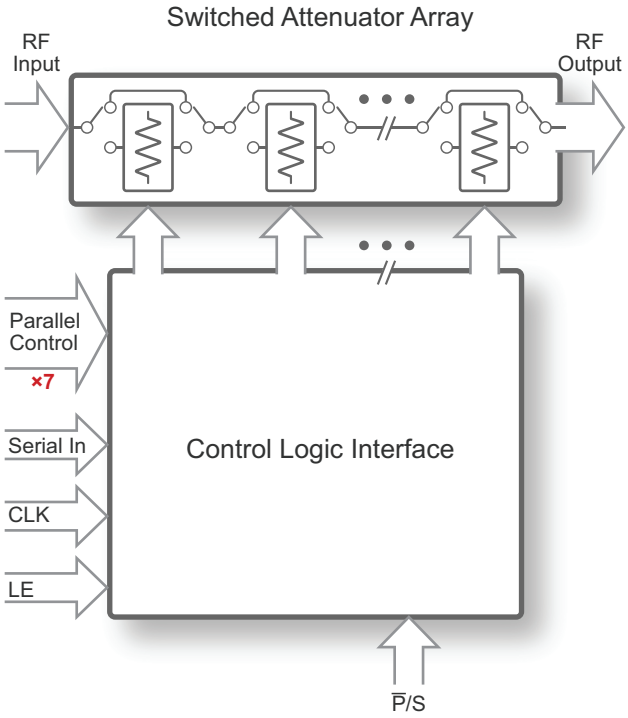
Features

- Flexible attenuation steps of 0.25 dB, 0.5 dB, and 1 dB up to 31.75 dB
- Glitch-less attenuation state transitions
- Monotonicity: 0.25 dB up to 4 GHz, 0.5 dB up to 5 GHz, and 1 dB up to 6 GHz
- Operating temperature: Extended +125 °C
- Parallel and serial programming interfaces
- Bidirectional functionality
- Packaging: 24-lead 4 × 4 mm QFN

Applications

- 3G/4G wireless infrastructure
- Land mobile radio (LMR) systems
- Point-to-point communication systems

Figure 1 • PE43711 functional diagram



Product description

The PE43711 is a 50Ω, HaRP™ technology-enhanced, 7-bit RF digital step attenuator (DSA) that supports a broad frequency range from 9 kHz to 6 GHz. It features glitch-less attenuation state transitions and supports a 1.8V control voltage and an extended operating temperature range to +125 °C, making this device ideal for many broadband wireless applications.

The PE43711 is a pin-compatible and upgraded version of the PE43502, PE43503, PE43602, and PE43702. An integrated digital control interface supports both serial and parallel programming of the attenuation, including the capability to program an initial attenuation state at power-up.

The PE43711 covers a 31.75-dB attenuation range in 0.25-dB, 0.5-dB, and 1-dB steps. It can maintain 0.25-dB monotonicity through 4 GHz, 0.5-dB monotonicity through 5 GHz, and 1-dB monotonicity through 6 GHz. In addition, no external blocking capacitors are required if 0 VDC is present on the RF ports.

The PE43711 is manufactured on pSemi's UltraCMOS® process, a patented variation of silicon-on-insulator (SOI) technology on a sapphire substrate. pSemi's HaRP technology enhancements deliver high linearity and excellent harmonics performance. It is an innovative feature of the UltraCMOS process.

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Absolute maximum ratings

Exceeding the absolute maximum ratings listed in **Table 1** could cause permanent damage. Restrict operation to the limits in **Table 2**. Operation between the operating range maximum and the absolute maximum for extended periods could reduce reliability.

ESD precautions

When handling this UltraCMOS device, observe the same precautions as with any other ESD-sensitive devices. Although this device contains circuitry to protect it from damage due to ESD, do not exceed the rating specified in **Table 1**.

Latch-up immunity

Unlike conventional CMOS devices, UltraCMOS devices are immune to latch-up.

Table 1 • PE43711 absolute maximum ratings

Parameter or condition	Min	Max	Unit
Supply voltage, V_{DD}	–0.3	5.5	V
Digital input voltage	–0.3	3.6	V
RF input power, 50Ω: 9 kHz–48 MHz >48 MHz–6 GHz	–	**Figure 5** +31	 dBm dBm
Storage temperature range	–65	+150	°C
ESD voltage HBM, all pins[1]	–	3000	V
ESD voltage CDM, all pins[2]	–	1000	V

Notes:
1) Human body model (MIL–STD 883 Method 3015).
2) Charged device model (JEDEC JESD22-C101).

Recommended operating conditions

Table 2 lists the PE43711 recommending operating conditions. Do not operate devices outside the recommended operating conditions listed below.

Table 2 • PE43711 recommended operating conditions

Parameter	Min	Typ	Max	Unit
Supply voltage, V_{DD}	2.3	–	5.5	V
Supply current, I_{DD}	–	150	200	µA
Digital input high	1.17	–	3.6	V
Digital input low	–0.3	–	0.6	V
Digital input current	–	–	17.5	µA
RF input power, CW:(1) 9 kHz–48 MHz >48 MHz–6 GHz	–	–	**Figure 5** +23	 dBm dBm
RF input power, pulsed:(2) 9 kHz–48 MHz >48 MHz–6 GHz	–	–	**Figure 5** +28	 dBm dBm
Operating temperature range	–40	+25	+125	°C

Notes:
1) 100% duty cycle, all bands, 50Ω.
2) Pulsed, 5% duty cycle of 4620-µs period, 50Ω.

Electrical specifications

Table 3 lists the PE43711 key electrical specifications at 25 °C, V_{DD} = 3.3V, RF1 = RF_{IN}, RF2 = RF_{OUT} (Z_S = Z_L = 50Ω), unless otherwise specified.

Table 3 • PE43711 electrical specifications

Parameter	Condition	Frequency	Min	Typ	Max	Unit
Operating frequency	–	–	9 kHz		6 GHz	As shown
Attenuation range	0.25-dB step 0.5-dB step 1-dB step	–	–	0–31.75 0–31.50 0–31.00	–	dB dB dB
Insertion loss	–	9 kHz–1.0 GHz 1.0–2.2 GHz 2.2–4.0 GHz 4.0–6.0 GHz	–	1.3 1.6 1.9 2.4	1.5 1.85 2.4 2.8	dB dB dB dB
Attenuation error	**0.25-dB step**					
	0–31.75 dB	9 kHz–2.2 GHz	–	–	±(0.15 + 1.5% of attenuation setting)	dB
	0–31.75 dB	>2.2–3.0 GHz	–	–	± (0.15 + 2.5% of attenuation setting)	dB
	0–31.75 dB	>3.0–4.0 GHz	–	–	± (0.25 + 3.5% of attenuation setting)	dB
	0.5-dB step					
	0–31.5 dB	9 kHz–2.2 GHz	–	–	± (0.15 + 1.5% of attenuation setting)	dB
	0–31.5 dB	>2.2–3.0 GHz	–	–	± (0.15 + 2.5% of attenuation setting)	dB
	0–31.5 dB	>3.0–5.0 GHz	–	–	± (0.25 + 3.5% of attenuation setting)	dB
	1-dB step					
	0–31 dB	9 kHz–2.2 GHz	–	–	± (0.15 + 1.5% of attenuation setting)	dB
	0–31 dB	>2.2–3.0 GHz	–	–	± (0.15 + 2.5% of attenuation setting)	dB
	0–31 dB	>3.0–5.0 GHz	–	–	± (0.25 + 3.5% of attenuation setting)	dB
	0–31 dB	>5.0–6.0 GHz	–	–	± (0.25 + 6.0% of attenuation setting)	dB
Return loss	Input port or output port	9 kHz–4 GHz 4–6 GHz	–	14 16	–	dB dB

Table 3 • PE43711 electrical specifications (cont.)

Parameter	Condition	Frequency	Min	Typ	Max	Unit
Relative phase	All states	9 kHz–4 GHz 4–6 GHz	–	31 48	–	deg deg
Input 0.1-dB compression point(*)	–	48 MHz–6 GHz	–	31	–	dBm
Input IP3	Two tones at +18 dBm and 20- MHz spacing	4 GHz 6 GHz	–	57 56	–	dBm dBm
RF T_{rise}/T_{fall}	10%/90% RF	–	–	200	–	ns
Settling time	RF settled to within 0.05 dB of the final value	–	–	1.6	–	µs
Switching time	50% CTRL to 90% or 10% RF	–	–	275	–	ns
Attenuation transient (envelope)	–	2 GHz	–	0.3	–	dB

Note: * The input 0.1-dB compression point is a linearity figure of merit. For the operating RF input power (50Ω), see **Table 2**.

Spurious performance

The PE43711 typical spurious performance is -165 dBm/Hz. The spur fundamental occurs around 8.5 MHz.

Glitch-less attenuation state transitions

The PE43711 features a novel architecture to provide the best-in-class glitch-less transition behavior when changing attenuation states. When RF input power is applied, the output power spikes are greatly reduced (≤0.3 dB) during attenuation state changes as compared to previous-generation DSAs.

Truth tables

Table 4 and **Table 5** list the PE43711 truth tables.

Table 4 • Parallel truth table

Parallel control setting							Attenuation setting RF1–RF2
D6	D5	D4	D3	D2	D1	D0	
L	L	L	L	L	L	L	Reference IL
L	L	L	L	L	L	H	0.25 dB
L	L	L	L	L	H	L	0.5 dB
L	L	L	L	H	L	L	1 dB
L	L	L	H	L	L	L	2 dB
L	L	H	L	L	L	L	4 dB
L	H	L	L	L	L	L	8 dB
H	L	L	L	L	L	L	16 dB
H	H	H	H	H	H	H	31.75 dB

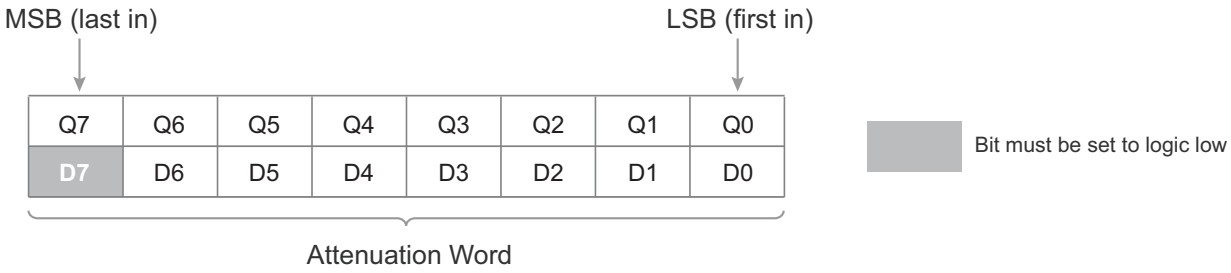
Table 5 • Serial attenuation word truth table

Attenuation word								Attenuation setting RF1–RF2
D7	D6	D5	D4	D3	D2	D1	D0 (LSB)	
L	L	L	L	L	L	L	L	Reference IL
L	L	L	L	L	L	L	H	0.25 dB
L	L	L	L	L	L	H	L	0.5 dB
L	L	L	L	L	H	L	L	1 dB
L	L	L	L	H	L	L	L	2 dB
L	L	L	H	L	L	L	L	4 dB
L	L	H	L	L	L	L	L	8 dB
L	H	L	L	L	L	L	L	16 dB
L	H	H	H	H	H	H	H	31.75 dB

Serial register map

Figure 2 shows the PE43711 serial register map.

Figure 2 • Serial register map



The attenuation word is derived directly from the value of the attenuation state. To find the attenuation word, multiply the value of the state by four, then convert to binary.

For example, to program the 12.5 dB state:

$$4 \times 12.5 = 50$$
$$50 \rightarrow 00110010$$

Serial Input: 00110010

Programming options

Parallel/serial selection

You can use a parallel or serial interface to control the PE43711. The $\overline{P}$/S bit provides this selection:

- $\overline{P}$/S = LOW selects the parallel interface.
- $\overline{P}$/S = HIGH selects the serial interface.

Parallel mode interface

The parallel interface consists of seven CMOS-compatible control lines that select the attenuation state, as listed in **Table 4**.

The parallel interface timing requirements are shown in **Figure 4** (Parallel Interface Timing Diagram), **Table 8** (Parallel and Direct Interface AC Characteristics) and switching time (**Table 3**).

For latched parallel programming, hold the Latch Enable (LE) LOW while changing the attenuation state control values, then pulse LE HIGH to LOW (per **Figure 4**) to latch the new attenuation state into the device.

For direct parallel programming, pull the LE line HIGH. Changing the attenuation state control values change the device state to the new attenuation. Direct mode is ideal for manual control of the device using hardwire, switches, or jumpers.

Serial interface

The serial interface is an 8-bit serial-in, parallel-out shift register buffered by a transparent latch. The eight bits make up the attenuation word that controls the DSA. **Figure 3** shows an example timing diagram for programming a state.

The serial interface is controlled using three CMOS-compatible signals: SI, Clock (CLK), and LE. The SI and CLK inputs allow data to be serially entered into the shift register. Serial data is clocked in LSB first.

To prevent the attenuator value from changing as the data is entered, load the shift register while holding LE LOW. To latch the new data into the DSA, toggle the LE input HIGH and then LOW again.

- **Table 5** lists the attenuation word truth table.
- **Figure 2** shows a serial register programming example.
- **Figure 3** shows the serial timing diagram.

When using the DSA in serial mode, ground all parallel control inputs.

Power-up control settings

The PE43711 always initializes to the maximum attenuation setting (31.75 dB) on power-up for both serial and latched parallel modes of operation and remains in this setting until you latch in the next programming word. In direct parallel mode, the DSA can be preset to any state within the 31.75 dB range by presetting the parallel control pins before power-up. In this mode, there is a 400-µs delay between the time the DSA is powered-up and the time the preferred state is set. During this power-up delay, the device attenuates to the maximum attenuation setting (31.75 dB) before defaulting to your defined state. If the control pins are left floating in this mode during power-up, the device defaults to the minimum attenuation setting (the insertion loss state).

Dynamic operation between serial and parallel programming modes is possible:

- If the DSA powers up in serial mode ($\overline{P}$/S= HIGH), all the parallel control inputs DI[6:0] must be set to logic LOW. Before toggling to parallel mode, program the DSA serially to ensure that D[7] is set to logic LOW.
- If the DSA powers up in latched or direct parallel mode, set all parallel pins DI[6:0] to logic LOW before toggling to serial mode ($\overline{P}$/S= HIGH), and held LOW until the DSA has been programmed serially to ensure bit D[7] is set to logic LOW.

The sequencing is only required once during power-up. After power-up, you can toggle the DSA between serial and parallel programming modes at will.

Figure 3 • Serial timing diagram

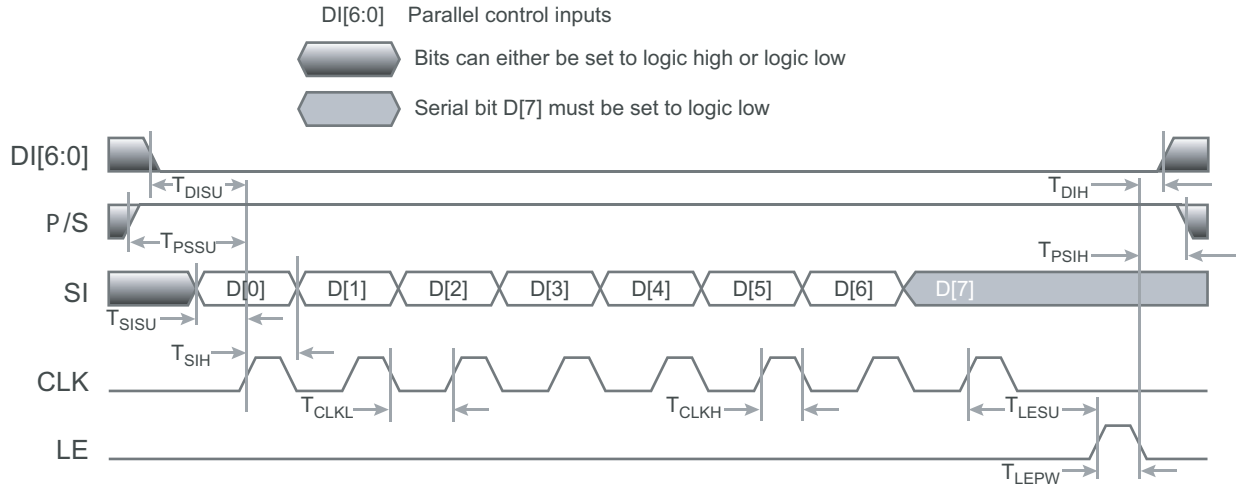


Figure 4 • Latched parallel/direct parallel timing diagram

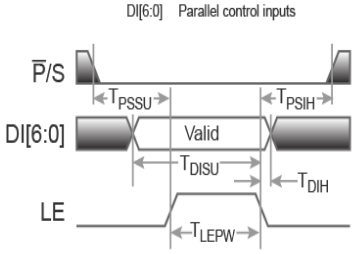


Table 6 • Latch and clock specification

Latch enable	Shift clock	Function
0	↑	Shift register clocked
↑	X	Contents of shift register transferred to attenuator core

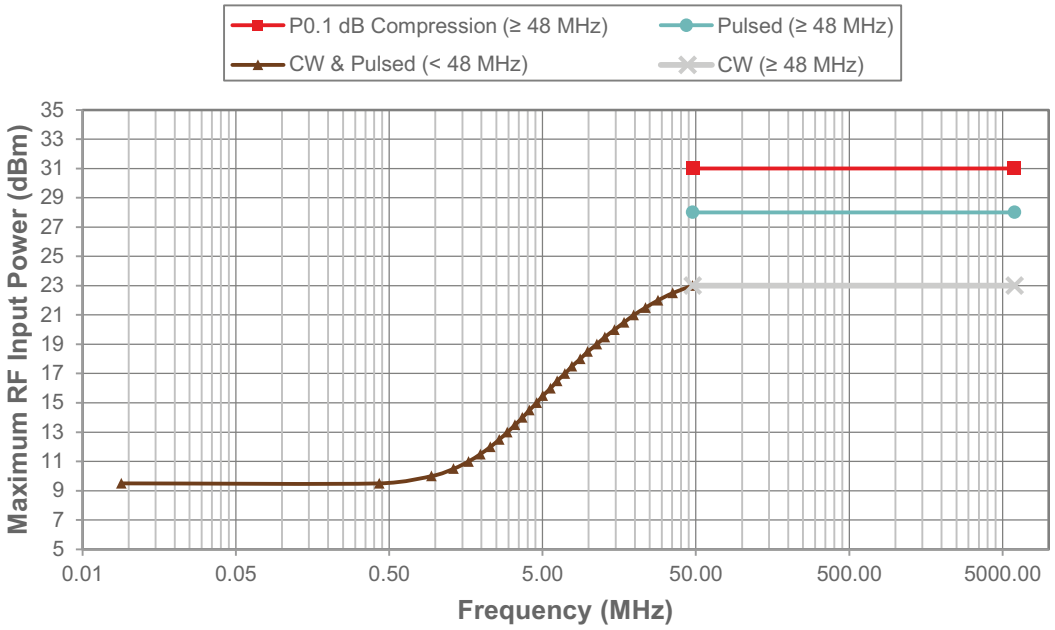
Table 7 • Serial interface AC characteristics(*)

Parameter or condition	Symbol	Min	Max	Unit
Serial clock frequency	F_{CLK}	–	10	MHz
Serial clock HIGH time	T_{CLKH}	30	–	ns
Serial clock LOW time	T_{CLKL}	30	–	ns
Last serial clock rising edge setup time to Latch Enable rising edge	T_{LESU}	10	–	ns
Latch Enable minimum pulse width	T_{LEPW}	30	–	ns
Serial data setup time	T_{SISU}	10	–	ns
Serial data hold time	T_{SIH}	10	–	ns
Parallel data setup time	T_{DISU}	100	–	ns
Parallel data hold time	T_{DIH}	100	–	ns
Address setup time	T_{ASU}	100	–	ns
Address hold time	T_{AH}	100	–	ns
Parallel/serial setup time	T_{PSSU}	100	–	ns
Parallel/serial hold time	T_{PSIH}	100	–	ns
Note: * V_{DD} = 3.3V or 5.0V, –40 °C < T_A < +125 °C, unless otherwise specified.				

Table 8 • Parallel and direct interface AC characteristics(*)

Parameter or condition	Symbol	Min	Max	Unit
Latch Enable minimum pulse width	T_{LEPW}	30	–	ns
Parallel data setup time	T_{DISU}	100	–	ns
Parallel data hold time	T_{DIH}	100	–	ns
Parallel/serial setup time	T_{PSSU}	100	–	ns
Parallel/serial hold time	T_{PSIH}	100	–	ns
Note: * V_{DD} = 3.3V or 5.0V, –40 °C < T_A < +125 °C, unless otherwise specified.				

Figure 5 • Power derating curve, 9 kHz–6 GHz, –40 to +125 °C ambient, 50Ω



Typical performance data

Figure 6–Figure 32 show the typical performance data at 25 °C and V_{DD} = 3.3V, RF1 = RF_{IN}, RF2 = RF_{OUT} (Z_S = Z_L = 50Ω), unless otherwise specified.

Figure 6 • Insertion loss vs. temperature

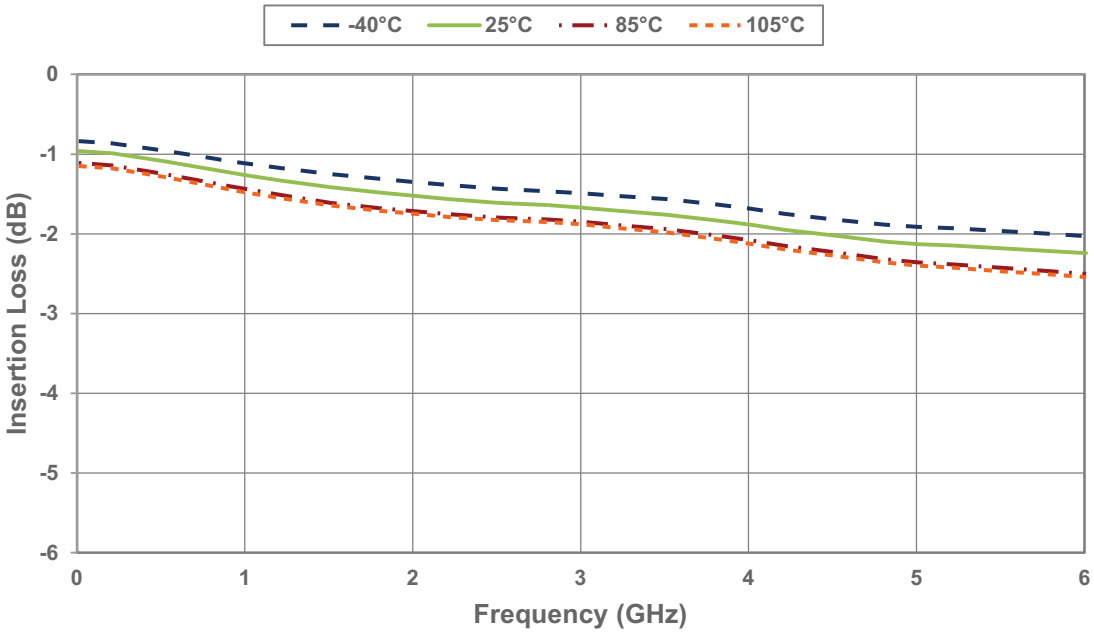


Figure 7 • Input return loss vs. attenuation setting

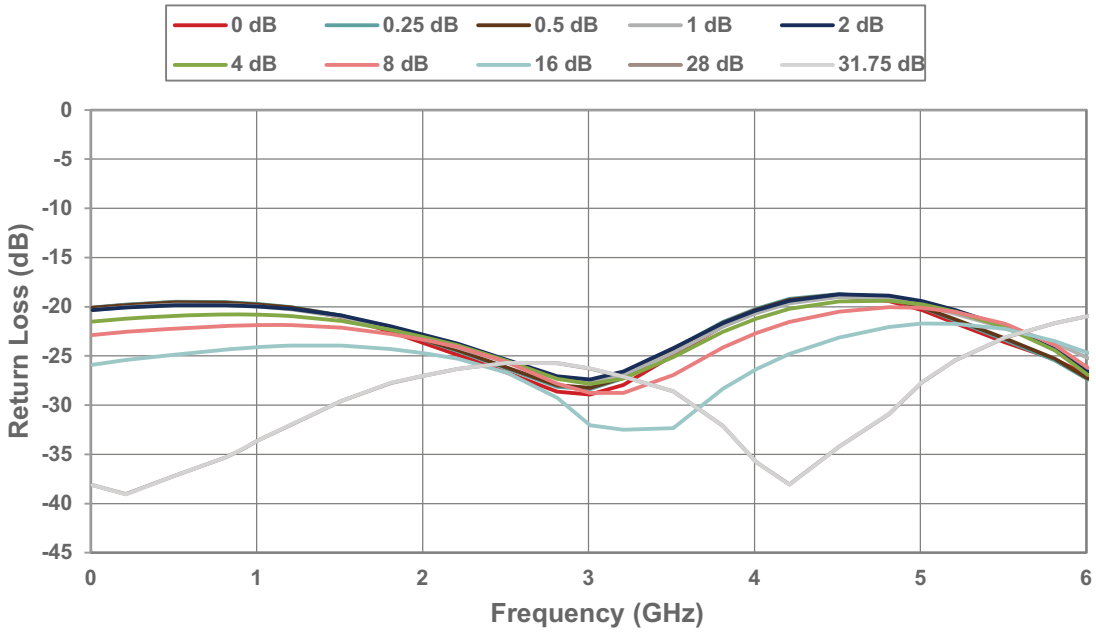


Figure 8 • Output return loss vs. attenuation setting

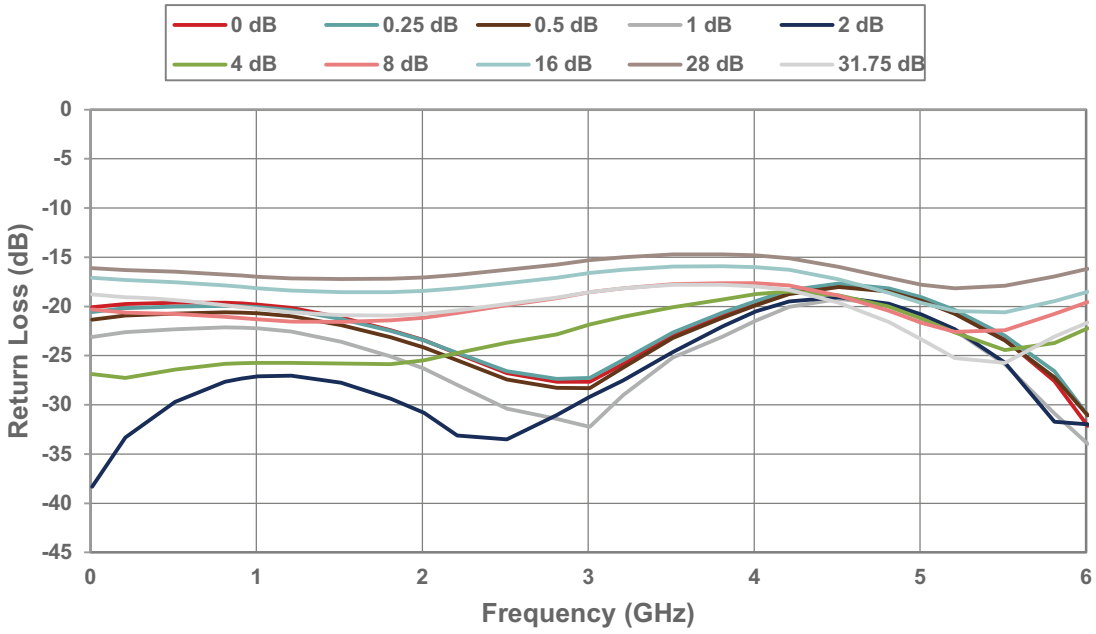


Figure 9 • Input return loss for 16 dB attenuation setting vs. temperature

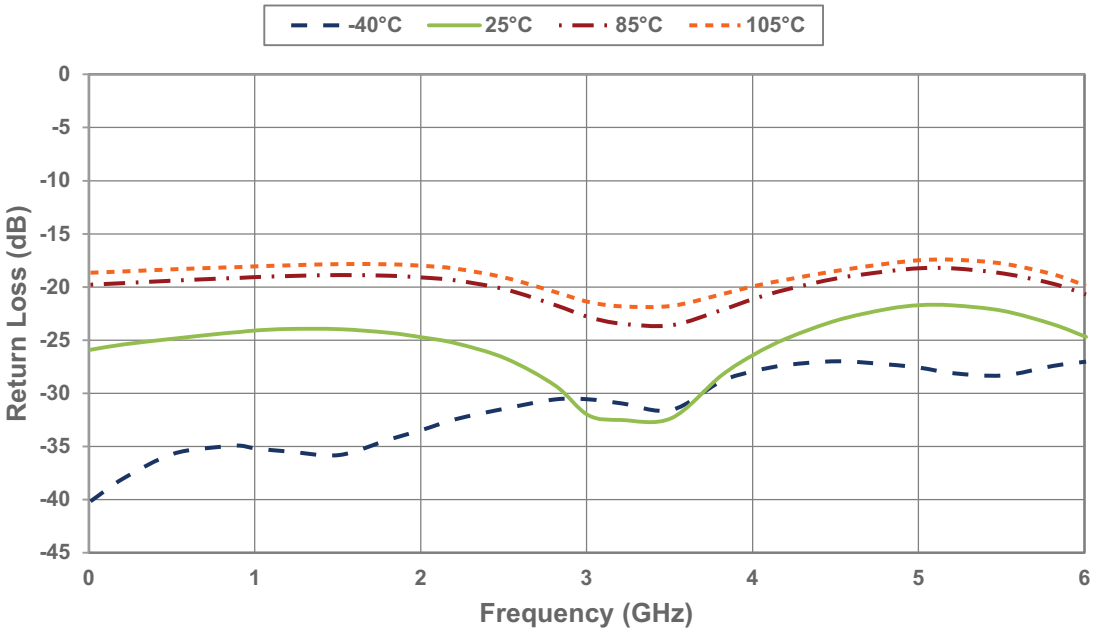


Figure 10 • Output return loss for 16 dB attenuation setting vs. temperature

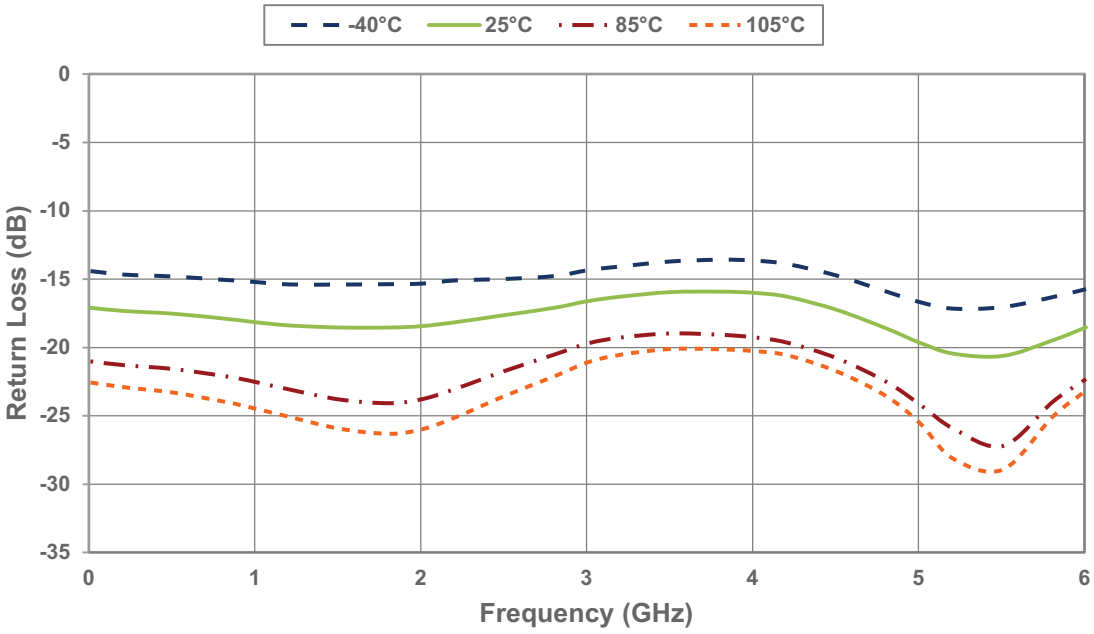


Figure 11 • Relative phase error vs. attenuation setting

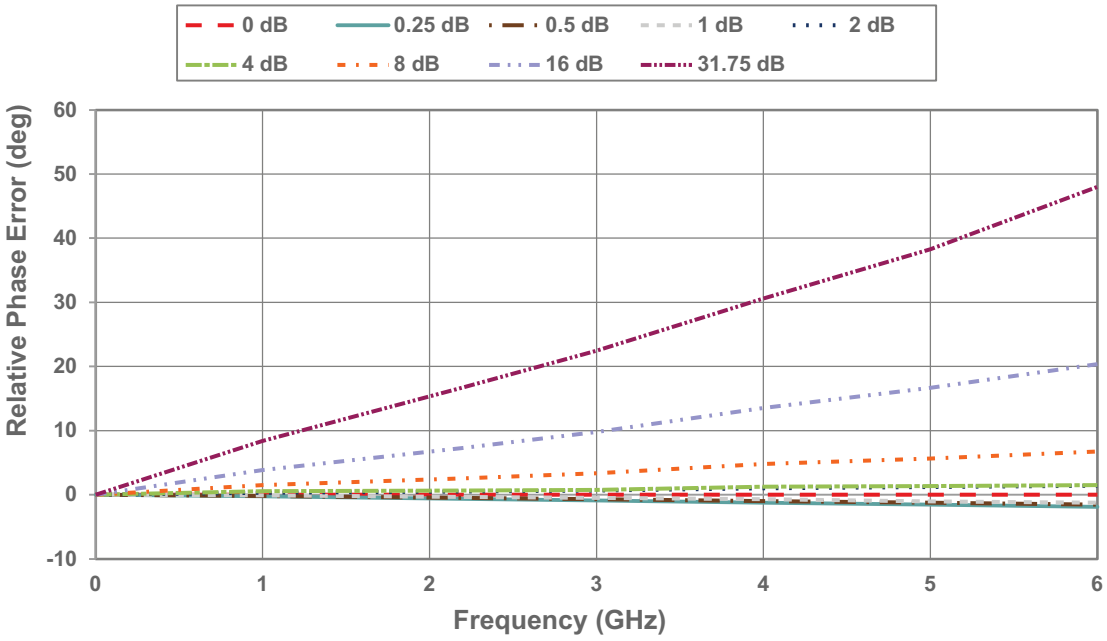


Figure 12 • Relative phase error for 31.75 dB attenuation setting vs. frequency

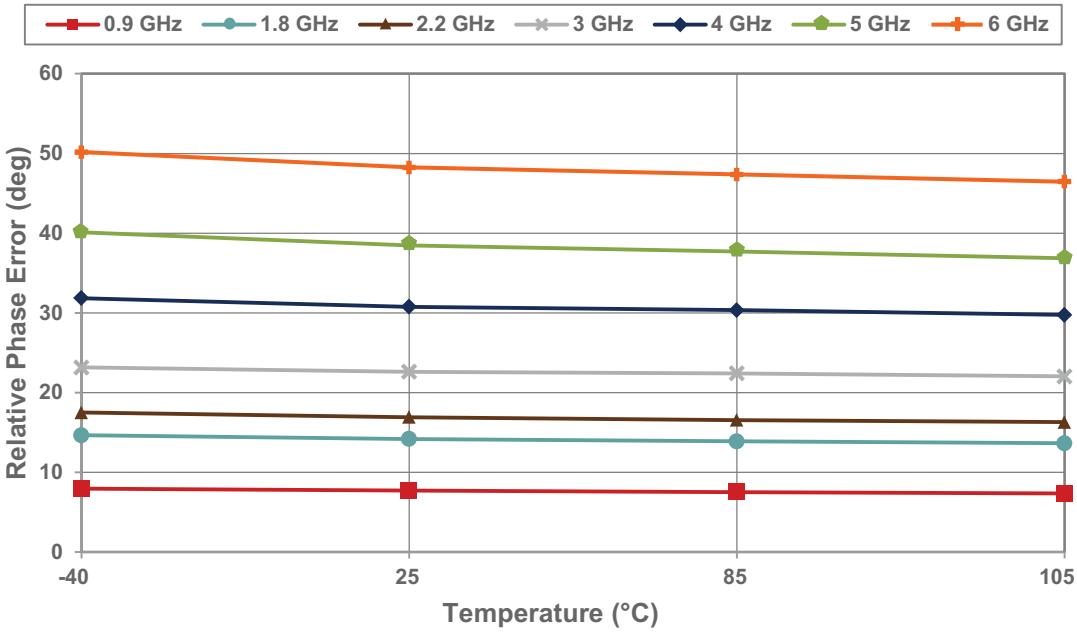


Figure 13 • Attenuation error at 900 MHz vs. temperature

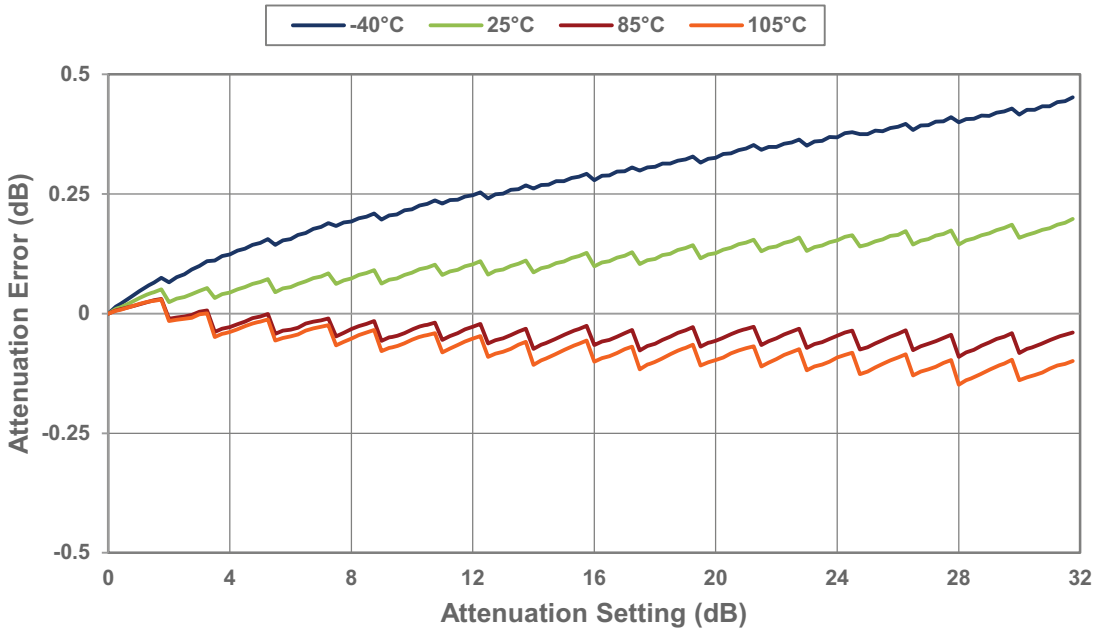


Figure 14 • Attenuation error at 1800 MHz vs. temperature

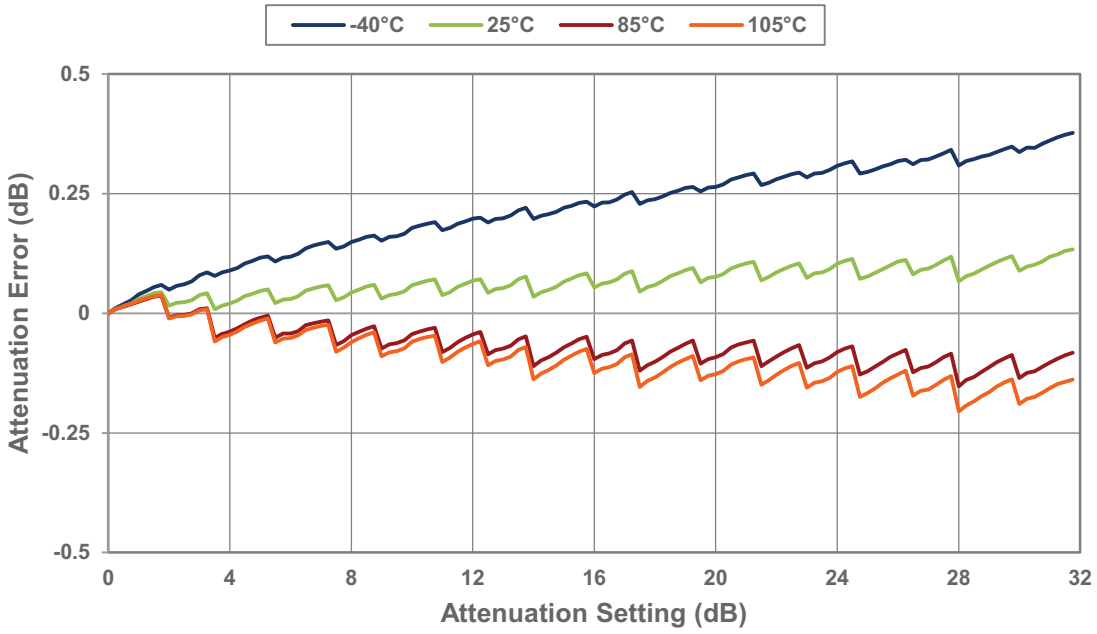


Figure 15 • Attenuation error at 2200 MHz vs. temperature

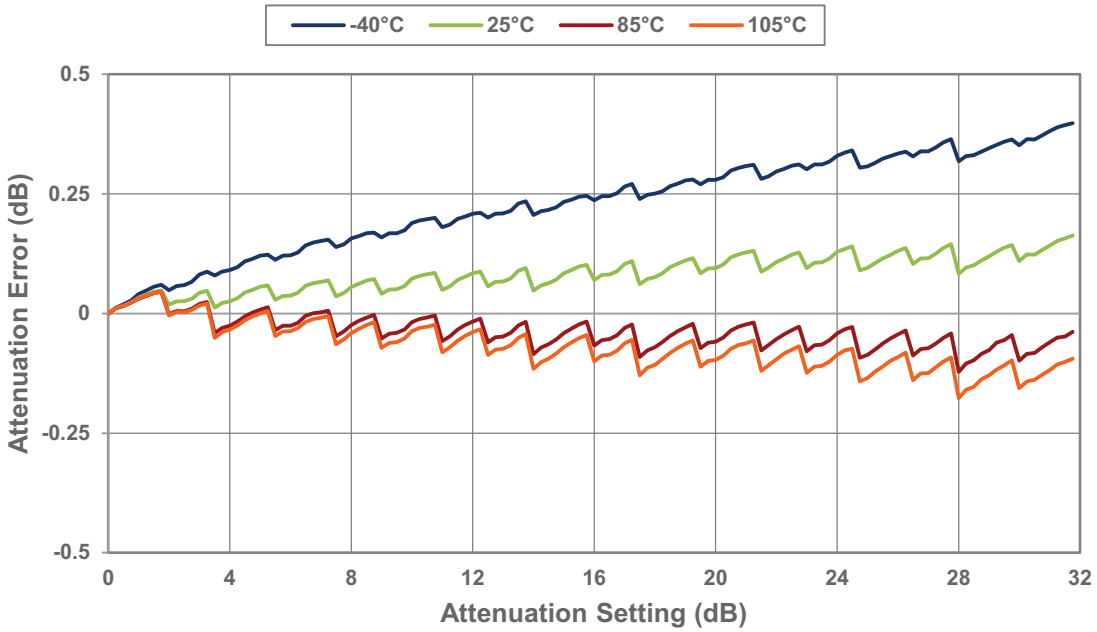


Figure 16 • Attenuation error at 3000 MHz vs. temperature

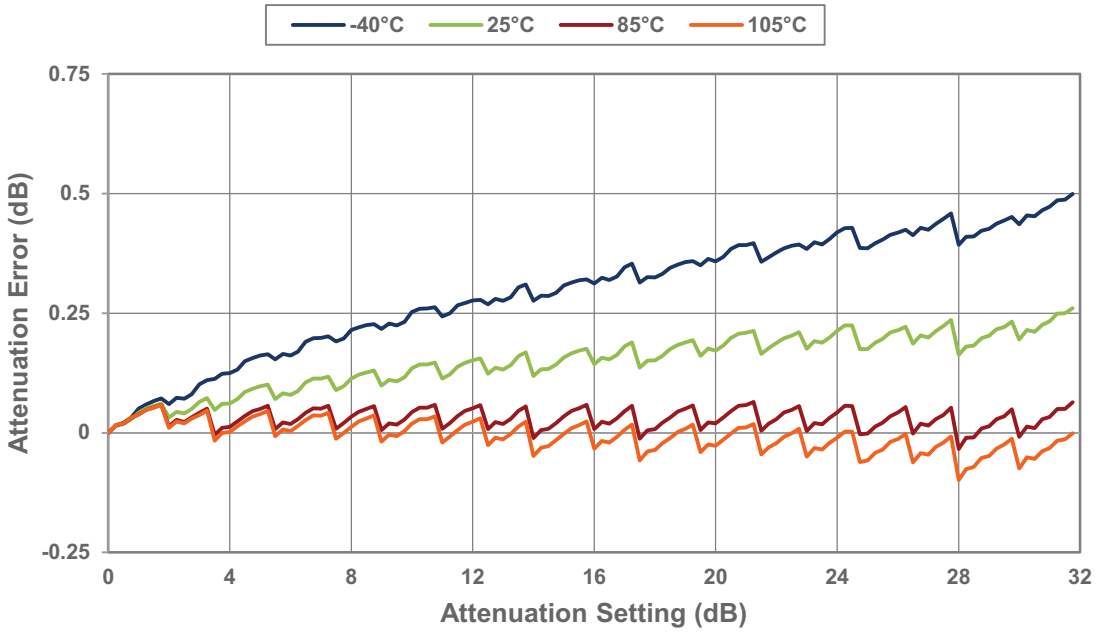


Figure 17 • Attenuation error at 4000 MHz vs. temperature

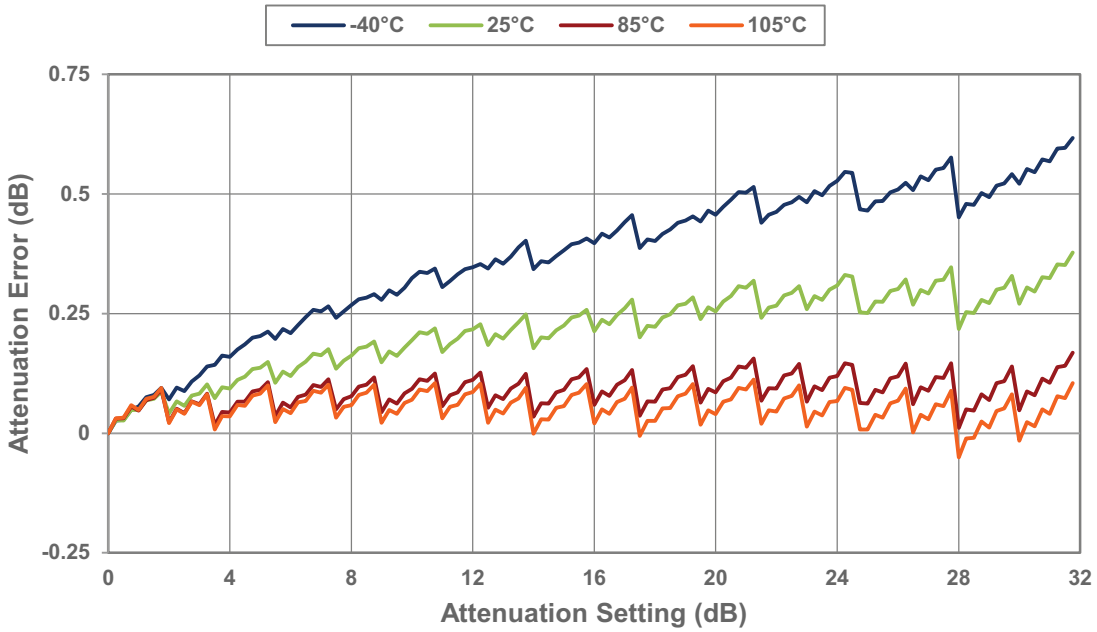


Figure 18 • IIP3 vs. attenuation setting

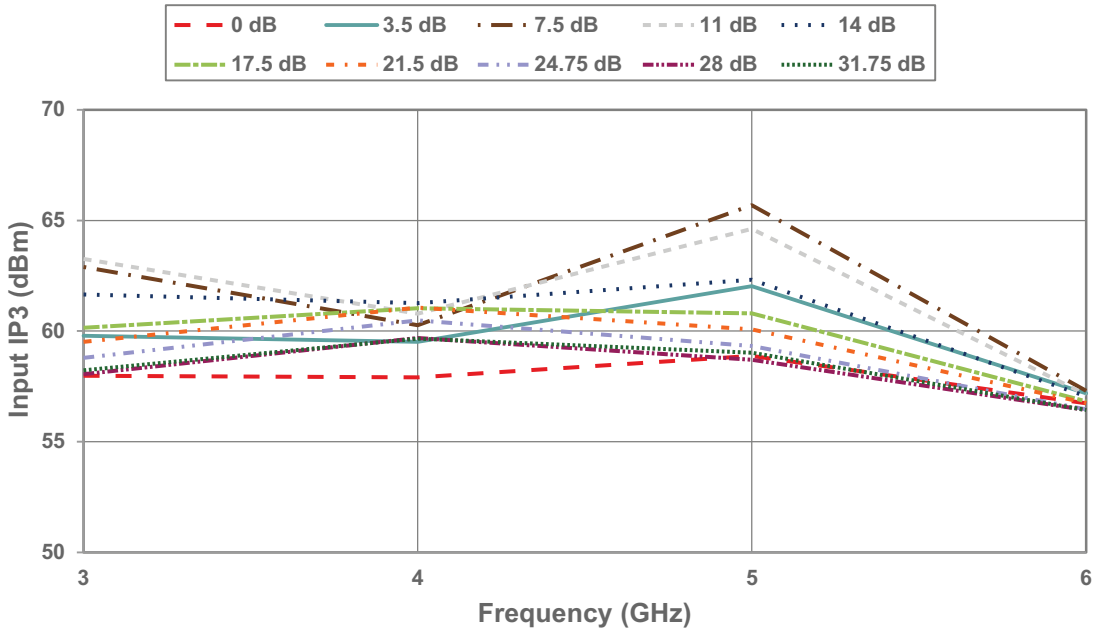
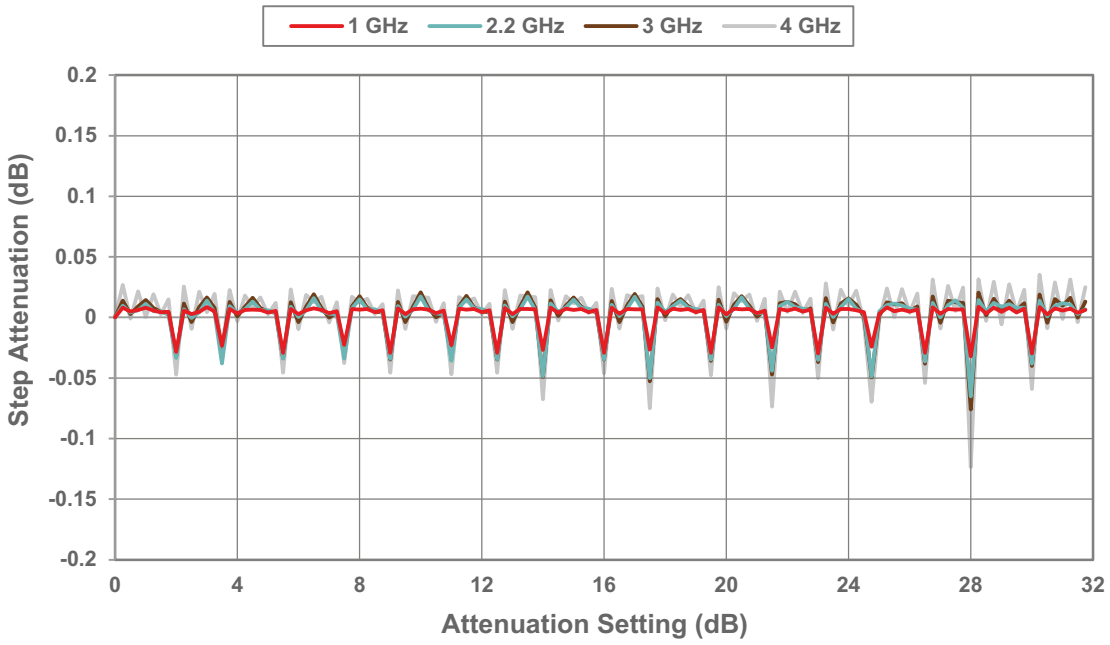


Figure 19 • 0.25 dB step attenuation vs. frequency(*)



Note: * Monotonicity is held so long as step attenuation does not cross below –0.25 dB.

Figure 20 • 0.25 dB step, actual vs. frequency

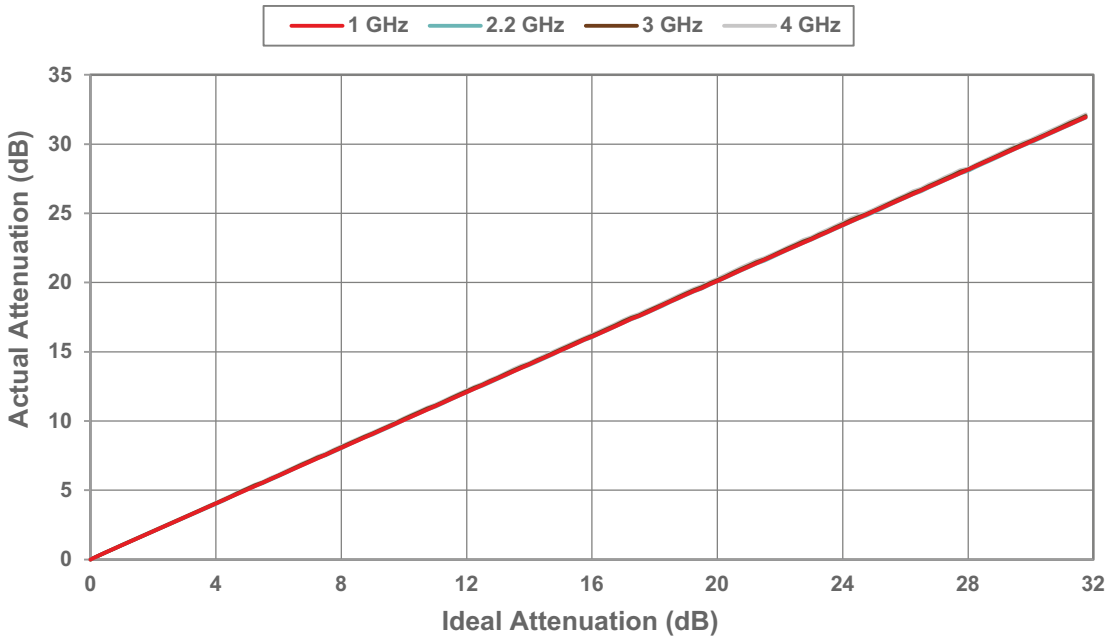


Figure 21 • 0.25 dB major state bit error vs. attenuation setting

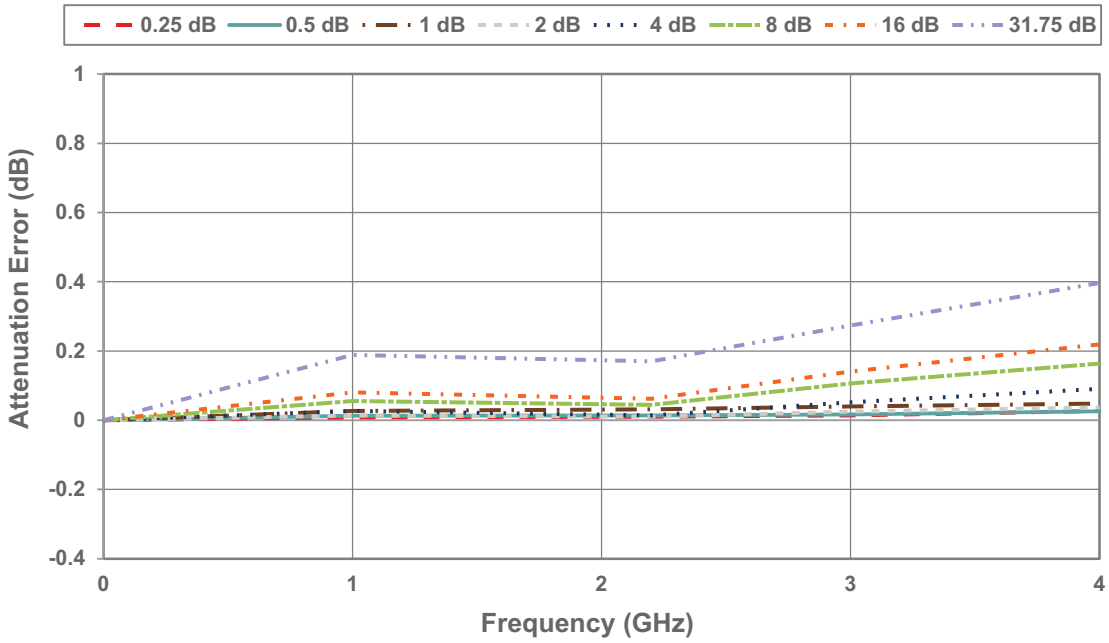


Figure 22 • 0.25 dB attenuation error vs. frequency

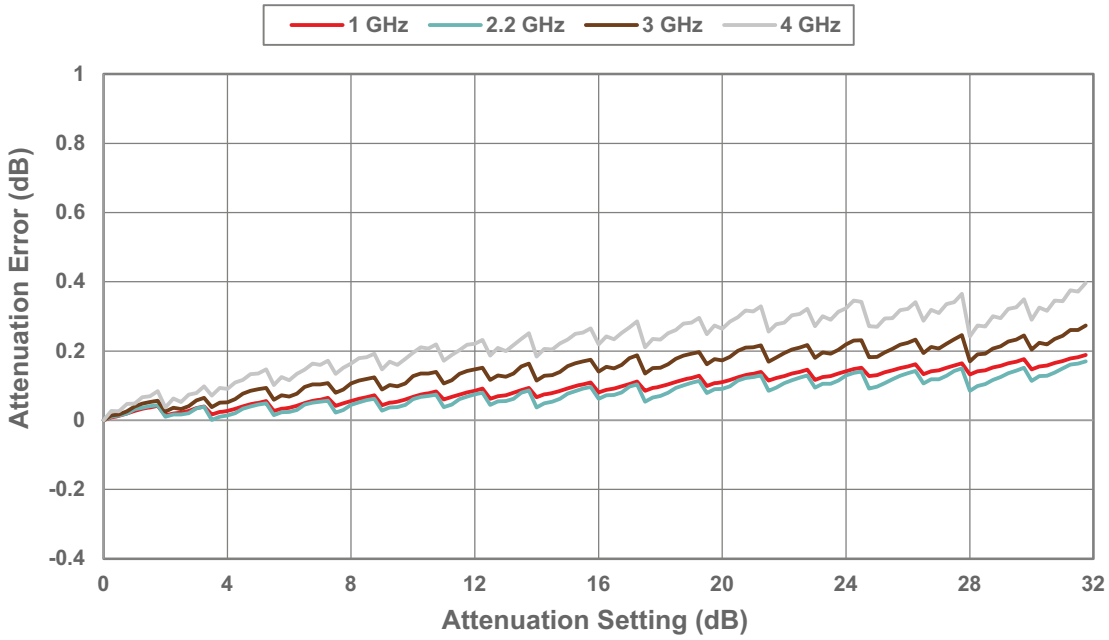
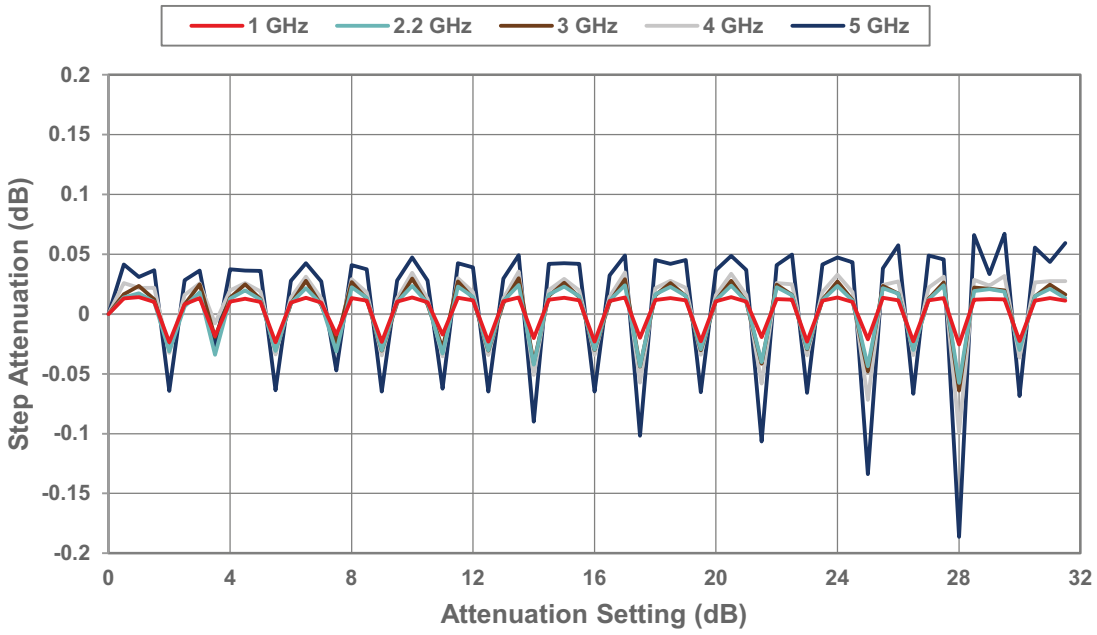


Figure 23 • 0.5 dB step attenuation vs. frequency(*)



Note: * Monotonicity is held so long as step attenuation does not cross below –0.5 dB.

Figure 24 • 0.5 dB step, actual vs. frequency

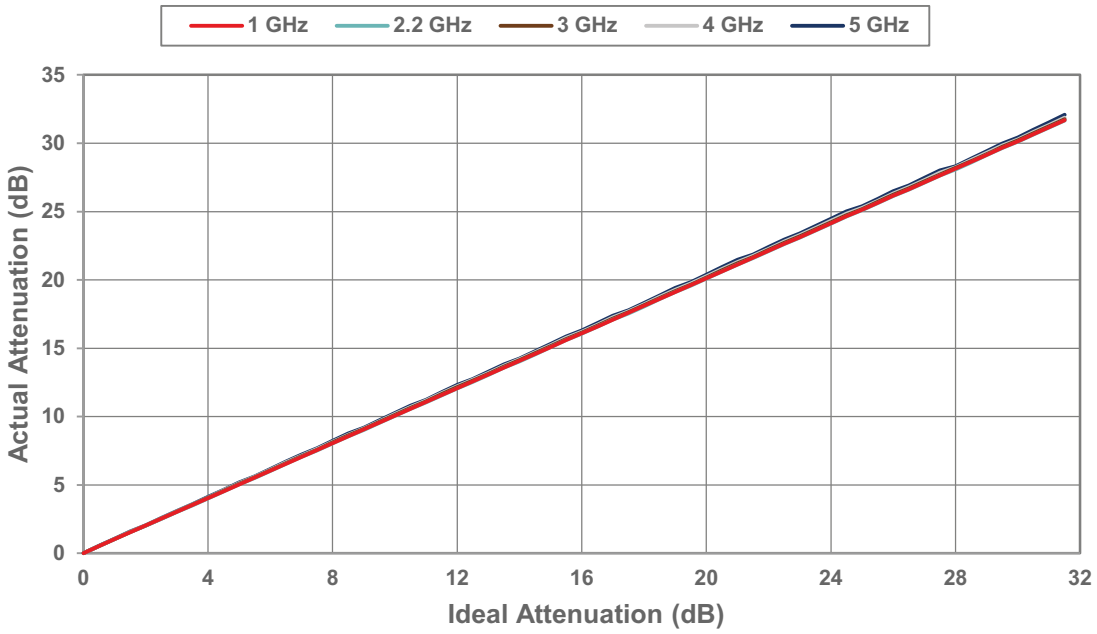


Figure 25 • 0.5 dB major state bit error vs. attenuation setting

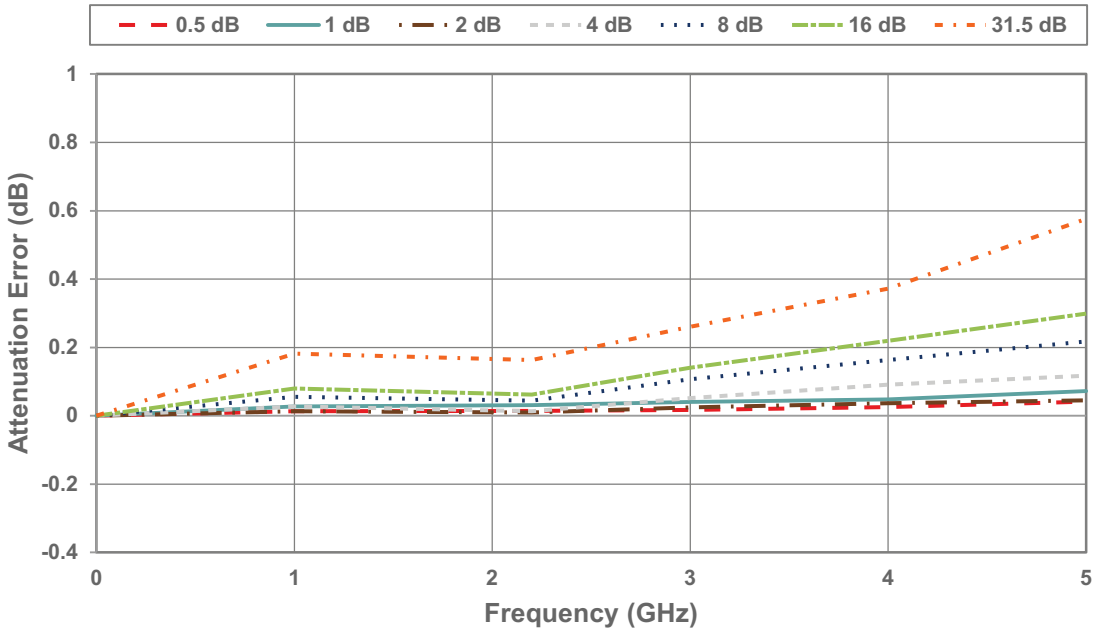


Figure 26 • 0.5 dB attenuation error vs. frequency

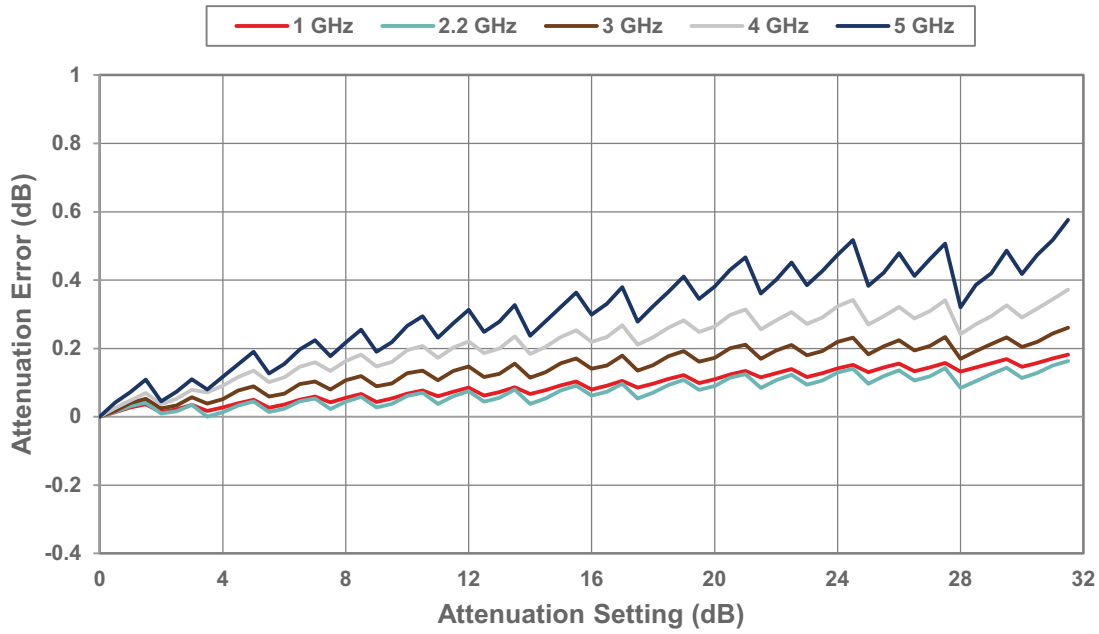
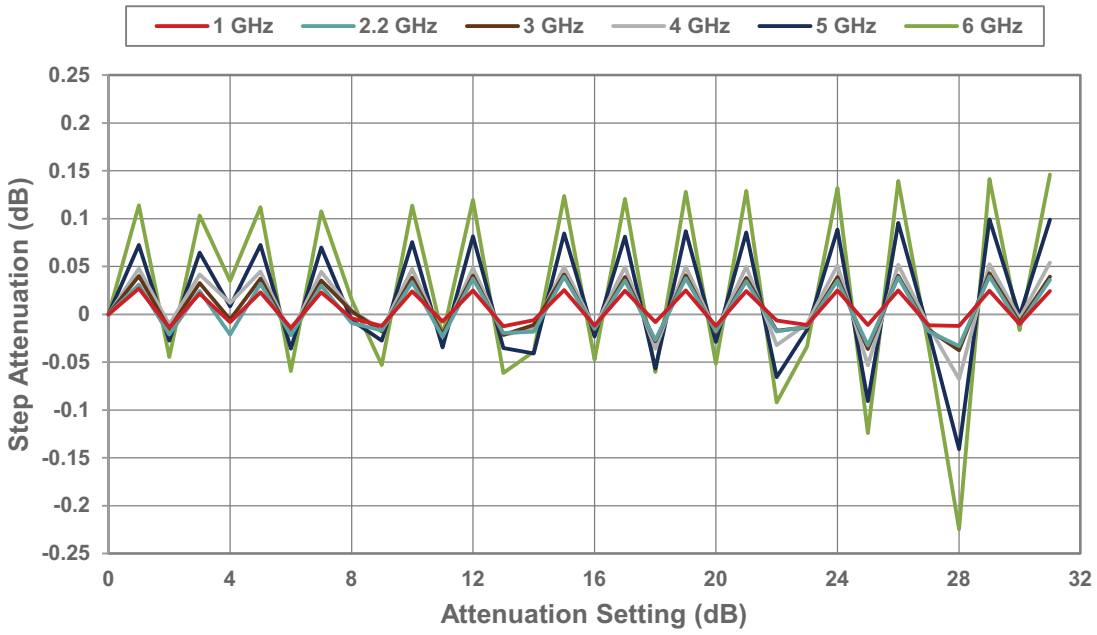


Figure 27 • 1 dB step attenuation vs. frequency(*)



Note: * Monotonicity is held so long as step attenuation does not cross below –1 dB.

Figure 28 • 1 dB step, actual vs. frequency

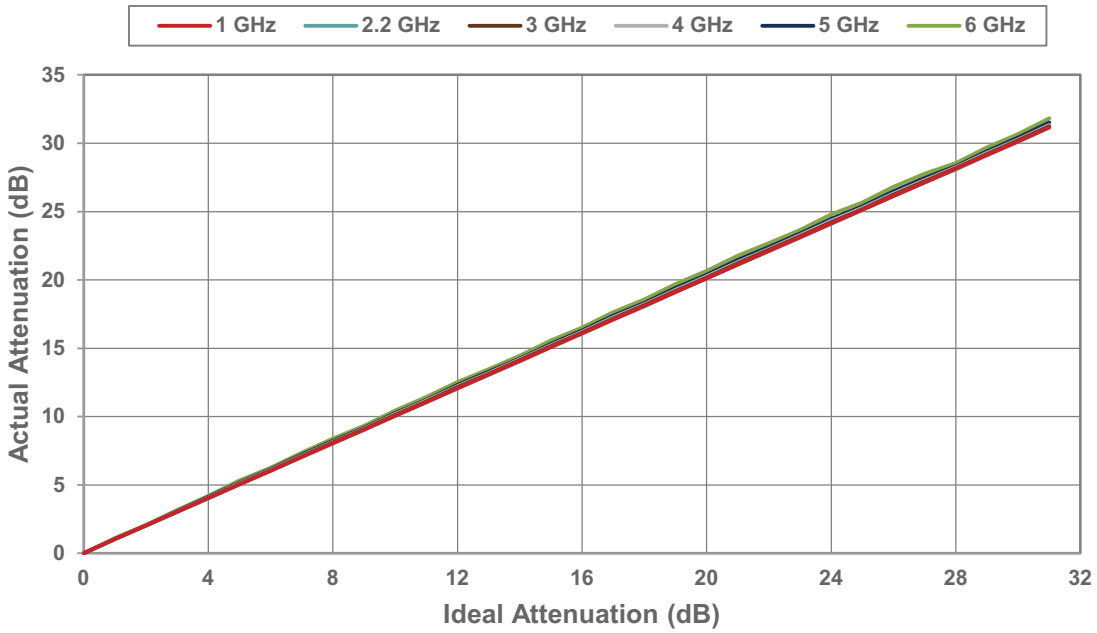


Figure 29 • 1 dB major state bit error vs. attenuation setting

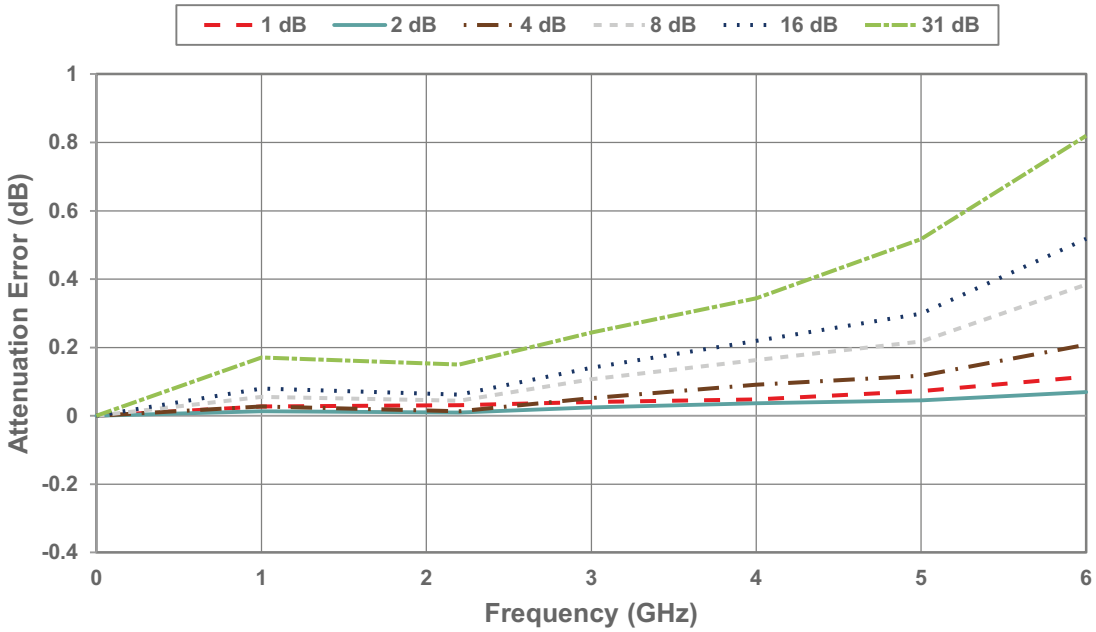


Figure 30 • 1 dB attenuation error vs. frequency

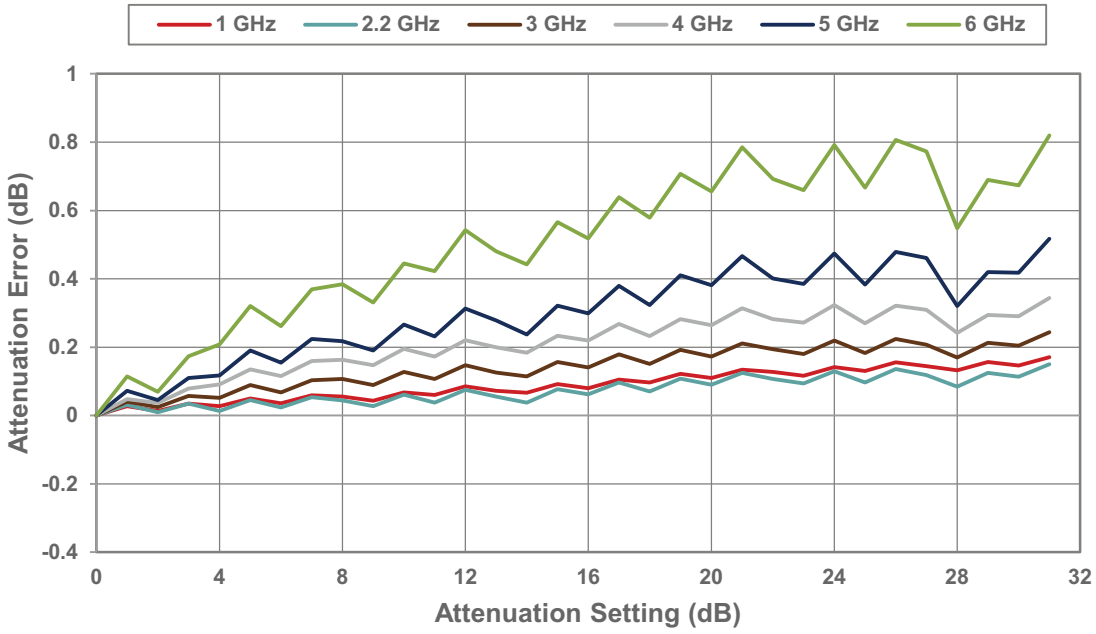


Figure 31 • Attenuation transient (15.75–16 dB), typical switching time = 275 ns

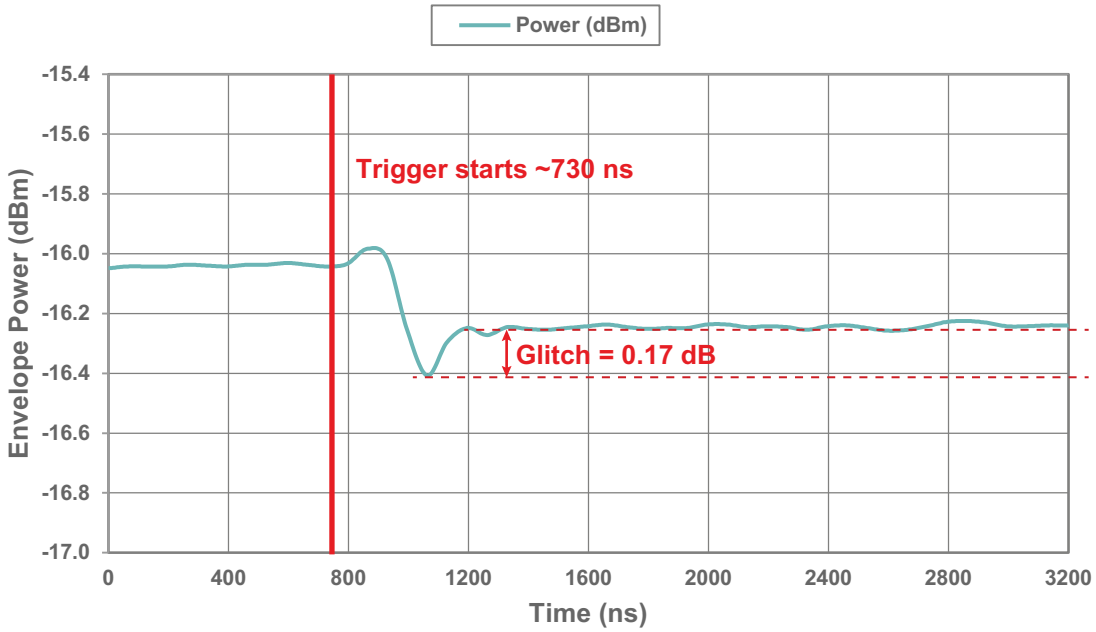
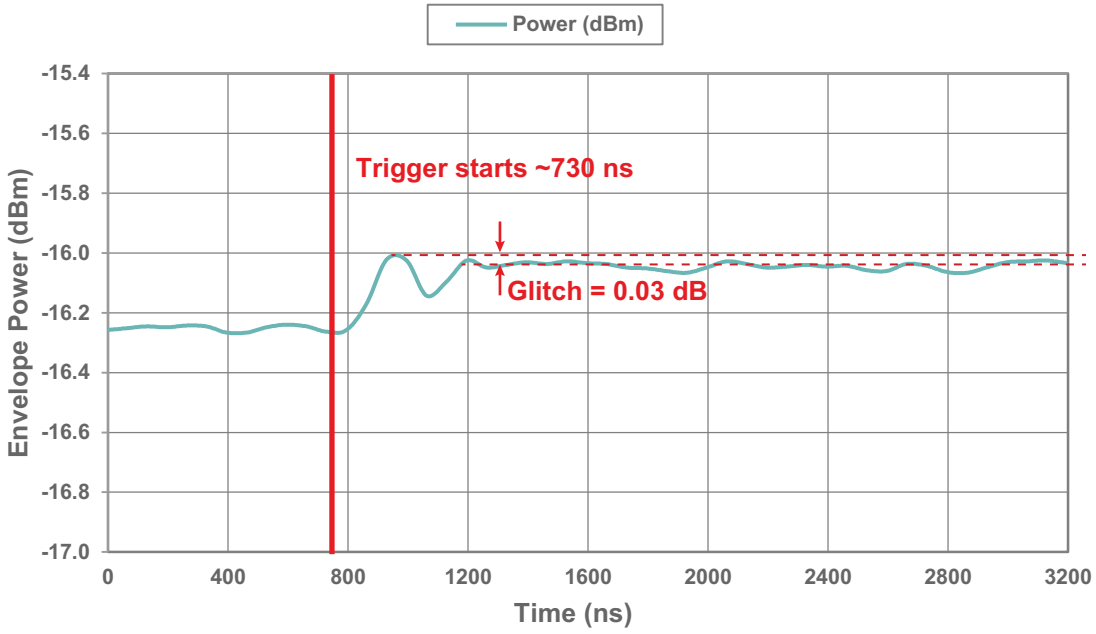


Figure 32 • Attenuation transient (16–15.75 dB), typical switching time = 275 ns



Evaluation kit

pSemi designed the digital step attenuator evaluation board (EVB) to ease your evaluation of the PE43711 digital step attenuator. The PE43711 EVB supports direct parallel, latched parallel, and serial modes.

Evaluation kit setup

Connect the EVB with the USB dongle board and the USB cable as shown in **Figure 33**.

Direct parallel programming procedure

Direct parallel programming is suitable for manual operation without software programming. For manual direct parallel programming, position the Parallel/ Serial (P/S) select switch to the Parallel position. Set the LE switch to the HIGH position. Switches D0–D6 are SP3T switches that allow you to manually program the parallel bits. When D0–D6 are toggled to the HIGH position, logic high is presented to the parallel input. When toggled to the LOW position, logic low is presented to the parallel input. Setting LE and D0–D6 to the EXT position presents as OPEN, which is set for software programming of latched parallel and serial modes. **Table 4** lists the parallel truth table.

Latched parallel programming procedure

For automated latched parallel programming, connect the USB dongle board and cable provided with the evaluation kit (EVK) from the USB port of the PC to the J5 header of the PE43711 EVB, and set the LE and D0–D6 SP3T switches to the EXT position. Position the Parallel/Serial (P/S) select switch to the parallel position. The evaluation software is written to operate the DSA in parallel mode. Verify that the software GUI is set to latched parallel mode. Use the software GUI to enable the preferred attenuation state. The software GUI automatically programs the DSA each time an attenuation state is enabled.

Serial programming procedure

For automated serial programming, connect the USB dongle board and cable provided with the EVK from the USB port of the PC to the J5 header of the PE43711 EVB, and set the LE and D0–D6 SP3T switches to the EXT position. Position the Parallel/ Serial (P/S) select switch to the serial position. The software GUI is written to operate the DSA in serial mode. Use the software GUI to enable each setting to the preferred attenuation state. The software GUI automatically programs the DSA each time an attenuation state is enabled.

Figure 33 • PE43711 evaluation kit

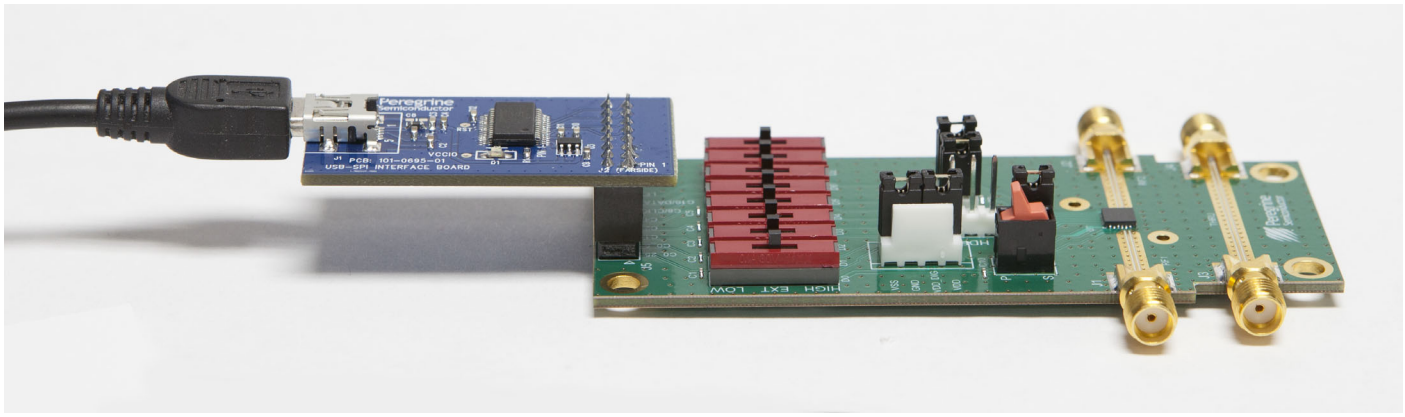
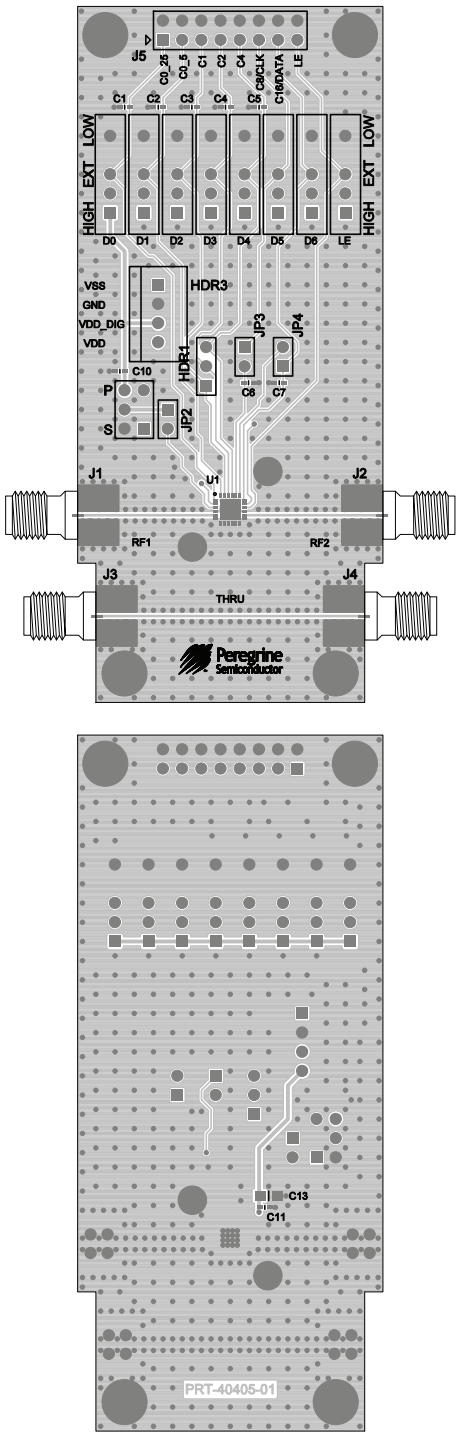


Figure 34 • PE43711 evaluation board layout



Pin configuration

Figure 35 shows the PE43711 pin map for the 24-lead 4 × 4 mm QFN package, and **Table 9** lists the description for each pin.

Figure 35 • Pin configuration, top view

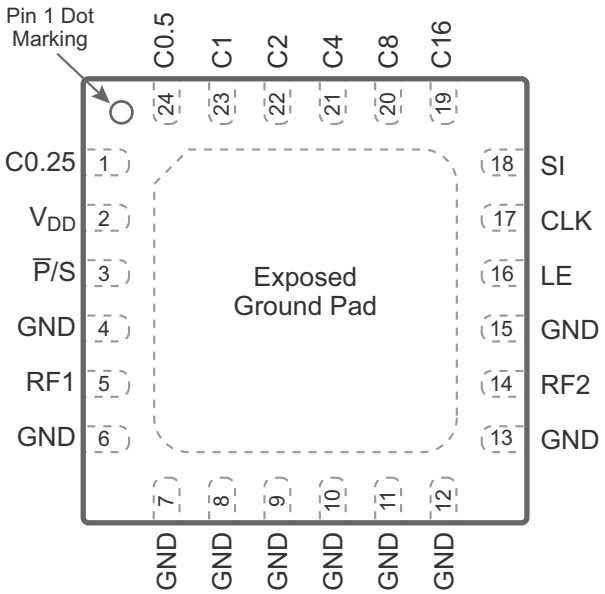


Table 9 • Pin descriptions

Pin no.	Pin name	Description
1	C0.25 (D0)[1]	Attenuation control bit, 0.25 dB
2	V_{DD}	Supply voltage
3	$\overline{P}$/S	Serial/parallel mode select
4, 6–13, 15	GND	Ground
5	RF1[2]	RF1 port (RF input)
14	RF2[2]	RF2 port (RF output)
16	LE	Serial interface Latch Enable input
17	CLK	Serial interface Clock input
18	SI	Serial interface Data input
19	C16 (D6)[1]	Parallel control bit, 16 dB
20	C8 (D5)[1]	Parallel control bit, 8 dB
21	C4 (D4)[1]	Parallel control bit, 4 dB
22	C2 (D3)[1]	Parallel control bit, 2 dB
23	C1 (D2)[1]	Parallel control bit, 1 dB
24	C0.5 (D1)[1]	Parallel control bit, 0.5 dB
Pad	GND	Exposed pad. Ground for proper operation.

Notes:

1) Ground C0.25, C0.5, C1, C2, C4, C8, and C16 if not in use.
2) RF pins 5 and 14 must be at 0 VDC. These RF pins do not require DC blocking capacitors for proper operation if the 0 VDC requirement is met.

Packaging information

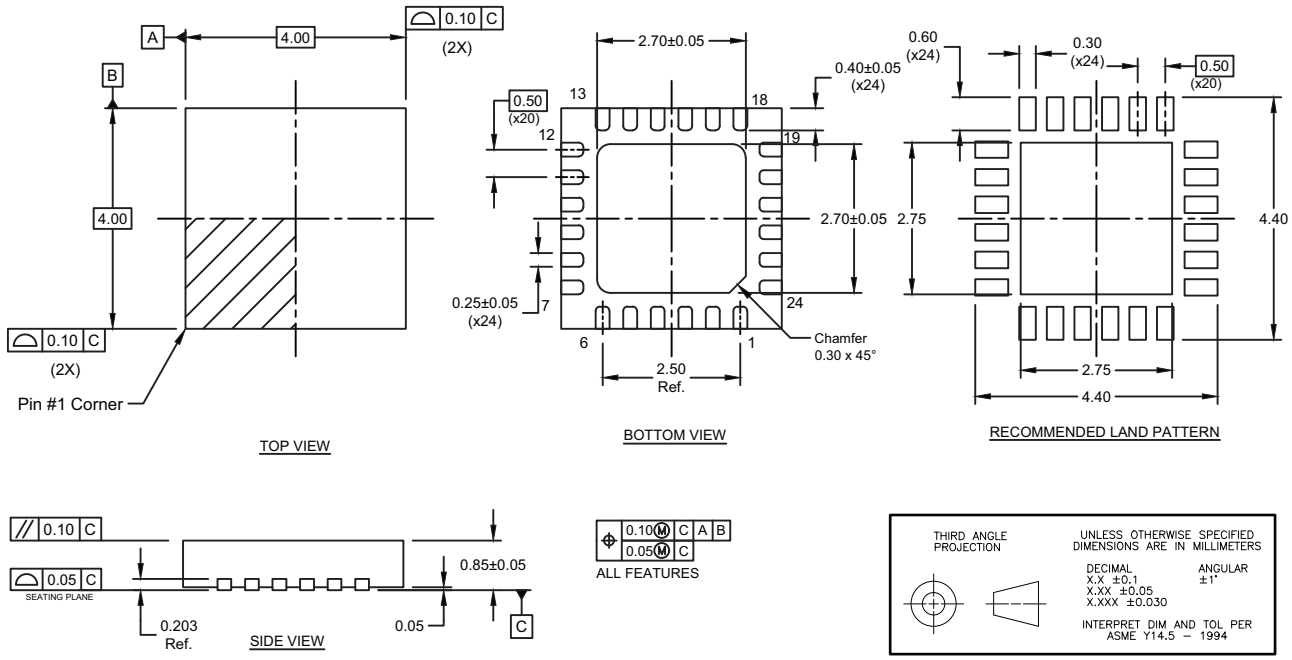
This section includes packaging data, including the moisture sensitivity level, package drawing, package marking, and tape-and-reel information.

Moisture sensitivity level

The PE43711 moisture sensitivity level rating in the 24-lead 4 × 4 mm QFN package is MSL1.

Package drawing

Figure 36 • Package mechanical drawing for the 24-lead 4 × 4 × 0.85 mm QFN package



Top-marking specification

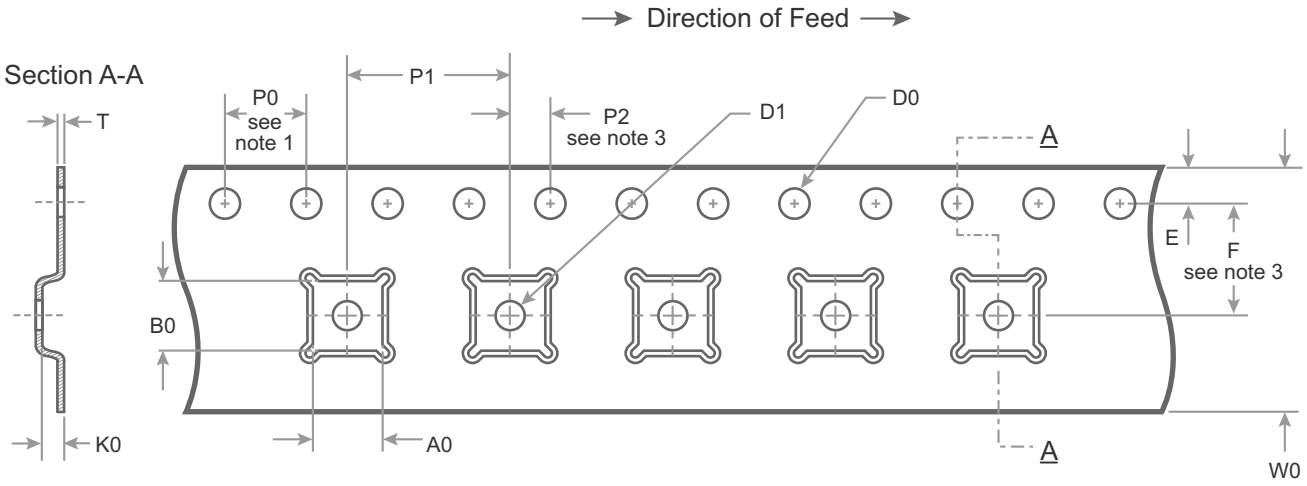
Figure 37 • PE43711 package marking specification



● = Pin 1 indicator
YY = Assembly year last two digits
WW = Assembly work week
ZZZZZZ = Assembly lot code

Tape and reel specification

Figure 38 • Tape and reel specification for the 24-lead 4 × 4 × 0.85 mm QFN package

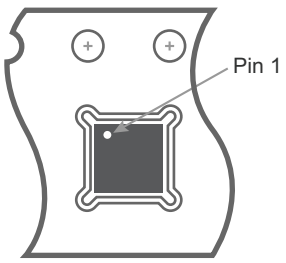


A0	4.35
B0	4.35
K0	1.10
D0	1.50 + 0.10/ -0.00
D1	1.50 min
E	1.75 ± 0.10
F	5.50 ± 0.05
P0	4.00
P1	8.00
P2	2.00 ± 0.05
T	0.30 ± 0.05
W0	12.00 ± 0.30

Notes:

1. 10 Sprocket hole pitch cumulative tolerance ±0.2
2. Camber in compliance with EIA 481
3. Pocket position relative to sprocket hole measured as true position of pocket, not pocket hole

Dimensions are in millimeters unless otherwise specified



Device Orientation in Tape

Ordering information

Table 10 lists the PE43711 order codes and shipping methods.

Table 10 • PE43711 order codes and shipping methods

Order code	Description	Packaging	Shipping method
PE43711B-Z	PE43711 digital step attenuator	Green 24-lead 4 × 4 mm QFN	3000 units/T&R
EK43711-03	PE43711 evaluation kit	Evaluation kit	1/box

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